

1. Administrative & Study Identification

Full Study Title: A Phase IIIb, Open-label, Multi-center Study to Evaluate Safety, Tolerability and Efficacy of OAV101 Administered Intrathecally to Participants With SMA Who Discontinued Treatment With Nusinersen or Risdiplam **Short Title/Acronym:** STRENGTH Study **Protocol Number:** COAV101B12301 (Internal Study ID) **NCT Number:** NCT05386680 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Intrathecal onasemnogene abeparvovec (OAV101 / Itvisma / Zolgensma IT) **Phase of Development:** Phase IIIb **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To characterize the safety and tolerability of intrathecal OAV101 (onasemnogene abeparvovec) in patients with Spinal Muscular Atrophy (SMA) aged 2 to <18 years who discontinued prior standard-of-care treatments. **Secondary Objectives:** To evaluate efficacy assessments including motor function (Hammersmith Functional Motor Scale-Expanded [HFMSE] and Revised Upper Limb Module [RULM]) and the caregiver experience (Assessment of Caregiver Experience in Neuromuscular Disease [ACEND]).

2.2 Background & Rationale To address an unmet medical need across the broader Spinal Muscular Atrophy (SMA) population by evaluating a one-time gene transfer therapy (OAV101 IT) for treatment-experienced patients. The study investigates safety and clinical stability/improvement in children and adolescents who have discontinued previous treatment with nusinersen (Spinraza) or risdiplam (Evrysdi).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 28 participants (Actual enrollment: 27 participants) **Number of Sites:** Multi-center, Global (Sites in the US, Australia, Belgium, Canada, France, Italy, Japan, Netherlands, Spain) **Estimated Study Start:** Mid-2022 **Estimated Primary Completion:** Late 2024 / Published 2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 2 to < 18 years at the time of dosing. 2. Confirmed diagnosis of Spinal Muscular Atrophy (SMA) with a documented SMA mutation. 3. Patients who are able to sit but are non-ambulatory (have never walked independently). 4. Discontinued prior standard-of-care SMA treatment (specifically, nusinersen or risdiplam).

4.2 Exclusion Criteria 1. Previous treatment with systemic gene therapy (e.g., intravenous onasemnogene AAV vector). 2. Any unstable disorder, severe respiratory distress, or contraindication to intrathecal puncture/administration. 3. Clinically significant anomalies in baseline laboratory parameters (e.g., severe hepatic impairment or severe thrombocytopenia).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: A single administration of 1.2×10^{14} vector genomes. **Route of Administration:** Intrathecal (IT) injection. **Duration of Treatment:** One-time dose followed by a 52-week observation/follow-up period. **Management Pathway:** Single-dose Intrathecal Administration Pathway.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive and multidisciplinary care for SMA as prescribed by the treating physician. **Prohibited:** Concurrent use of other SMA disease-modifying therapies (nusinersen and risdiplam must be thoroughly discontinued prior to enrollment).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -2	Informed Consent, Caregiver counseling on post-dosing observation and AESI signs, SMA Confirmation, Baseline Motor Assessments (HFMSE, RULM), Safety Labs
Baseline / Dosing	Day -1 to 1	Hospital admission (Day -1), Intrathecal Administration of OAV101 (Day 1), Acute AE monitoring, ACEND scale checklist review
Interim	Weeks 4, 12, 24, 36	Safety Labs, AESI monitoring, Motor Function Assessments, ACEND scale checklist review
End of Study	Week 52	Final HFMSE / RULM Assessments, Final Safety Review, ACEND scale checklist review, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Characterization of the safety and tolerability profile, measured by the incidence of Adverse

Events (AEs), Serious Adverse Events (SAEs), and Adverse Events of Special Interest (AESIs) up to 52 weeks post-dosing.

Measurement Method: Clinical assessment, physical exams, and standard Medical Dictionary for Regulatory Activities (MedDRA) queries (specifically monitoring for hepatotoxicity, transient thrombocytopenia, thrombotic microangiopathy, and dorsal root ganglia toxicity).

7.2 Secondary & Exploratory Endpoints * Change from baseline in Hammersmith Functional Motor Scale-Expanded (HFMSE) score at 52 weeks to assess motor function stabilization/improvement. * Change from baseline in the Revised Upper Limb Module (RULM) to assess motor function stabilization/improvement. * Assessment of Caregiver Experience in Neuromuscular Disease (ACEND), evaluated at every major clinical visit.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of any untoward medical occurrences volunteered by the participant/caregiver, detected via physical examination and laboratory findings, or identified through intermittent routine phone contacts for AE check-ins for up to 52 weeks. **Serious Adverse Events (SAEs):** Routine 24-hour reporting timeline. Common reported SAEs involved respiratory infections (e.g., pneumonia). **Data Monitoring Committee (DMC):** Independent committee for continuous monitoring of AESIs (e.g., hepatotoxicity, transient thrombocytopenia, and DRG toxicity).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients who received the intrathecal administration of OAV101). **Sample Size Justification:** Approximately 28 participants across 2 age brackets (2 to <6 years, and 6 to <18 years) to adequately characterize safety in a targeted, treatment-experienced pediatric population. **Handling of Missing Data:** Standard Mixed Model for Repeated Measures (MMRM) or Last Observation Carried Forward (LOCF) applied to efficacy scales (HFMSE, RULM) where applicable.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Rigorous onsite and remote monitoring corresponding with the 52-week clinical schedule, ensuring protocol adherence and proper AESI logging. **Education Protocol:** Mandatory specialized training for IT delivery, HFMSE clinical evaluator training, and DRG toxicity monitoring will be provided to all Healthcare Professionals (HCPs). **Audit Trail & Quality**

Mechanism: Standardized eCRF locking alongside a rigorous quality audit mechanism that includes centralized review of motor scale assessments (e.g., HFMSE scoring) and centralized lab toxicity screening.

11. Study Identifiers & Governance

Primary ID: {{NCT05386680}} **Registry Number:** {{NCT05386680}}
Sponsor/Lead Organization: {{Novartis Pharmaceuticals}} **Study Title:** {{STRENGTH Study}} **Official Regulatory Title:** {{A Phase IIb, Open-label, Multi-center Study to Evaluate Safety, Tolerability and Efficacy of OAV101 Administered Intrathecally to Participants With SMA Who Discontinued Treatment With Nusinersen or Risdiplam}}

12. Clinical Framework (SMA Specific)

Primary Condition: {{Spinal Muscular Atrophy (SMA)}} **Diagnosis Threshold:** {{Confirmed SMA mutation, able to sit but non-ambulatory, Age 2 to <18}} **Medical Methodology:** {{Intrathecal Gene Transfer Therapy}} **Optimization Target:** {{Motor function stabilization/improvement (HFMSE/RULM)}}

13. Study Procedures & Methodology

Management Pathway: {{Single-dose Intrathecal Administration Pathway}} **Education Protocol (HCP):** {{Specialized training for IT delivery, HFMSE clinical evaluator training, and DRG toxicity monitoring}} **Education Protocol (Patient):** {{Caregiver counseling on post-dosing observation and AESI signs}} **Quality Audit Mechanism:** {{Centralized review of motor scale assessments and centralized lab toxicity screening}}

14. Observation & Follow-Up Schedule

Total Duration: {{52 Follow-up weeks}} **Onsite Visit Cadence:** {{Baseline (Day -1/1), Weeks 4, 12, 24, 36, 52}} **Remote Monitoring Frequency:** {{Intermittent routine phone contacts for AE check-ins}} **Checklist Review Interval:** {{Every major clinical visit for ACEND scale}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				
Statistician	[Name]	_____	[DD-MMM-YYYY]				